

Project - Library Database

Learning Objectives

- Design a relational database schema for a library management system
- Write SQL queries to perform all CRUD operations on the library data
- Generate reports using aggregate functions and GROUP BY

Engage (5-7 minutes)

A school library manages thousands of books, borrowers, and transactions. Designing a proper database structure and writing queries to manage it all brings together everything learned in this chapter—from table design to advanced queries.

Explore (10-12 minutes)

In groups of three, design tables for Books, Members, and Borrowings. Identify primary keys, foreign keys, and appropriate data types. Sketch the relationships between tables on paper before writing any SQL.

Explain (10-12 minutes)

The library database has three core tables: Books (BookID, Title, Author, Category, Copies), Members (MemberID, Name, Grade, Phone), and Borrowings (BorrowID, BookID, MemberID, BorrowDate, ReturnDate). Borrowings links Books and Members through foreign keys. This normalization eliminates data redundancy and ensures consistency.

Elaborate (8-10 minutes)

Write complete SQL for: creating all three tables, inserting sample data (5 books, 5 members, 10 borrowings), and generating reports—most borrowed book, members with overdue books, and books never borrowed. Use JOIN concepts to combine data from multiple tables.

Evaluate (5-8 minutes)

Present your database design to the class. Submit a document with CREATE TABLE statements, sample INSERT statements, and five SELECT queries including at least one with JOIN, one with GROUP BY, and one with aggregate functions.